Vidhyadhan Charitable Trust

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Social Media Usage & Screen Time Analysis

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Introduction

- Social media is widely used for communication and entertainment.
- People spend more time on screens daily.
- Excessive screen time affects:
    1. Sleep quality
    2. Mental health
    3. Productivity
- This project analyzes user data to understa these effects.

Project Goals & Objectives

- Analyze dataset of **50,000 users.**
- Identify usage patterns across age and occupation.
- Study relation between:
    1. Screen time
    2. Stress levels
    3. Sleep duration
- Build ML models to:
    1. Predict phone usage
    2. Detect high stress users

Dataset – Overview

- Dataset file: **socialmedia_data.csv**
- Total Records: **50,000**

Includes:

- **Demographics:** Age, Gender, Occupation
- **Usage:** Daily phone hours, social media usage
- **Lifestyle:** Sleep hours, stress level
- **Device Type:** Android, iOS, PC

User_ID	Age	Gender	Occupation	Device_Type	Daily_Phone_Hours	Social_Media_Hours
U1	58	Male	Professional	Android	1.3	6.7
U2	25	Male	Professional	Android	1.2	1.5
U3	19	Male	Student	iOS	5.3	5.7
U4	35	Female	Business Owner	iOS	5.8	2.5
U5	33	Male	Freelancer	Android	7.9	1.3
U6	32	Female	Student	Android	10.9	4.2
U7	26	Male	Student	iOS	5.6	6.1
U8	24	Male	Freelancer	iOS	8.5	1.9
U9	52	Female	Business Owner	Android	9.4	2.5
U10	23	Female	Student	iOS	2.8	1.1
U11	55	Other	Professional	Android	8.3	1.1
U12	45	Other	Business Owner	Android	12	6.7
U13	20	Female	Business Owner	iOS	11.1	5.6
U14	19	Female	Freelancer	iOS	9.7	4.1
U15	23	Male	Professional	Android	1.1	2.5
U16	31	Male	Business Owner	Android	11.6	5.5
U17	32	Other	Professional	Android	2.3	7.8
U18	50	Female	Student	Android	10.8	5.7
U19	56	Female	Business Owner	Android	3.3	4.3
U20	19	Other	Professional	Android	3.9	4.2
U21	53	Other	Business Owner	Android	3.3	2.6
U22	30	Other	Professional	iOS	9.5	6.9
U23	59	Female	Freelancer	Android	10.9	4.9
U24	52	Female	Student	iOS	5.5	5.1
U25	44	Female	Business Owner	Android	3.7	4.2
U26	32	Female	Professional	iOS	10.1	7.5
U27	46	Male	Business Owner	iOS	1.2	6.5
U28	55	Other	Student	iOS	9.7	6.1
U29	35	Male	Professional	iOS	11.7	1.6
U30	18	Other	Student	iOS	11.4	7.8
U31	28	Other	Student	Android	2.1	2.3
U32	45	Other	Freelancer	Android	5.8	3.8
U33	39	Other	Professional	Android	8.2	2.3
U34	35	Other	Freelancer	iOS	3.4	2
U35	27	Female	Student	Android	11.8	6.3

Methodology & Tools

Python

Libraries Used:

- Pandas → Data cleaning & analysis.
- Matplotlib & Seaborn → Data visualization.
- Scikit-learn → Machine Learning.

Tools:

- VS Code
- Jupyter Notebook
- Google Colab

Data Cleaning & Preprocessing

- Removed missing values (**NULL**).
- Deleted duplicate records using **User_ID**.
- Converted data into proper format.
- Handled outliers for accurate results.

Feature Engineering

What is Feature Engineering?

- Feature Engineering is the process of **creating new features (columns)** from existing data.
- It helps improve **data analysis and machine learning performance**.
- Converts raw data into **meaningful insights**.

Benefits of Feature Engineering:

- Improves **model accuracy.**
- Makes data easier to understand.
- Helps find **hidden patterns.**
- Enables better **predictions.**

Estimated Weekly Screen Time

- A new feature to calculate **total weekly usage.**

Why this is useful?

- Gives a **complete view** of user screen time
- More accurate than just daily usage

Example:

- Daily = 4 hours
- Weekend = 6 hours
- Weekly = (4×5) + (6×2) = **32 hours**

Formula Used:

Weekly Screen Time = (Daily Hours × 5) + (Weekend Hours × 2)

High Stress Indicator (Binary Feature)

- Created a new column to identify **high stress. users.**

Condition:

- Stress Level ≥ 7 → **High Stress (Yes).**
- Stress Level < 7 → **Normal (No).**

Why this is useful?

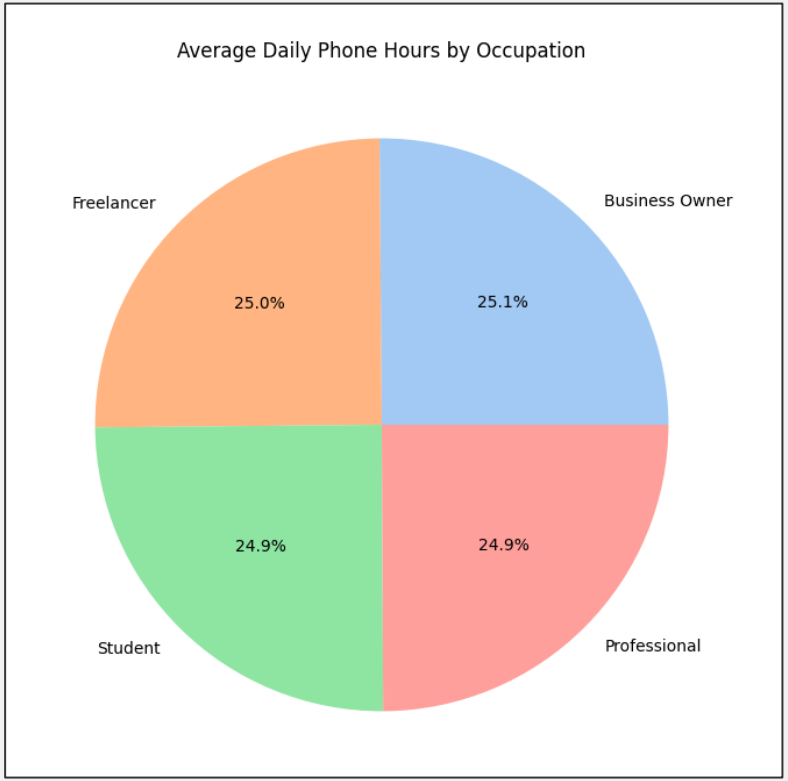
- Converts complex data into **simple categories**
- Helps in **classification model (ML)**

Example:

- Stress = 8 → High Stress.
- Stress = 5 → Normal

EDA 1 – Occupation vs Phone Usage

- Compared phone usage across occupations.
- Students and IT professionals use phones the most



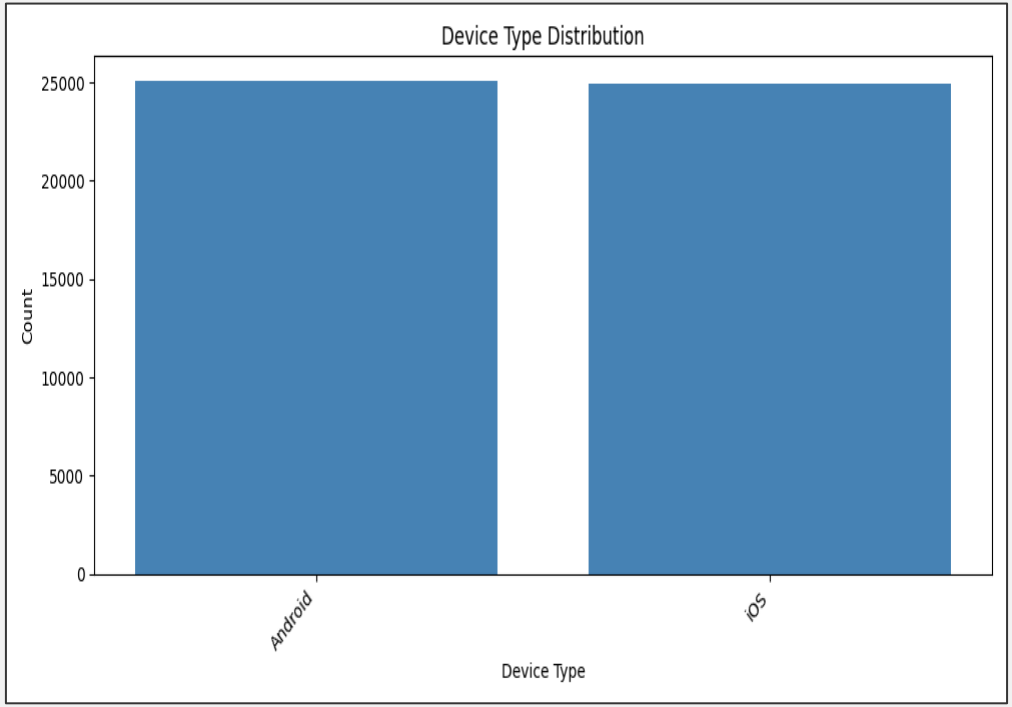
EDA 2 – Device & Age Trends

- **Device usage comparison:**

1. Android vs iOS vs PC.

- **Age-based trends:**

1. Young adults have highest usage.

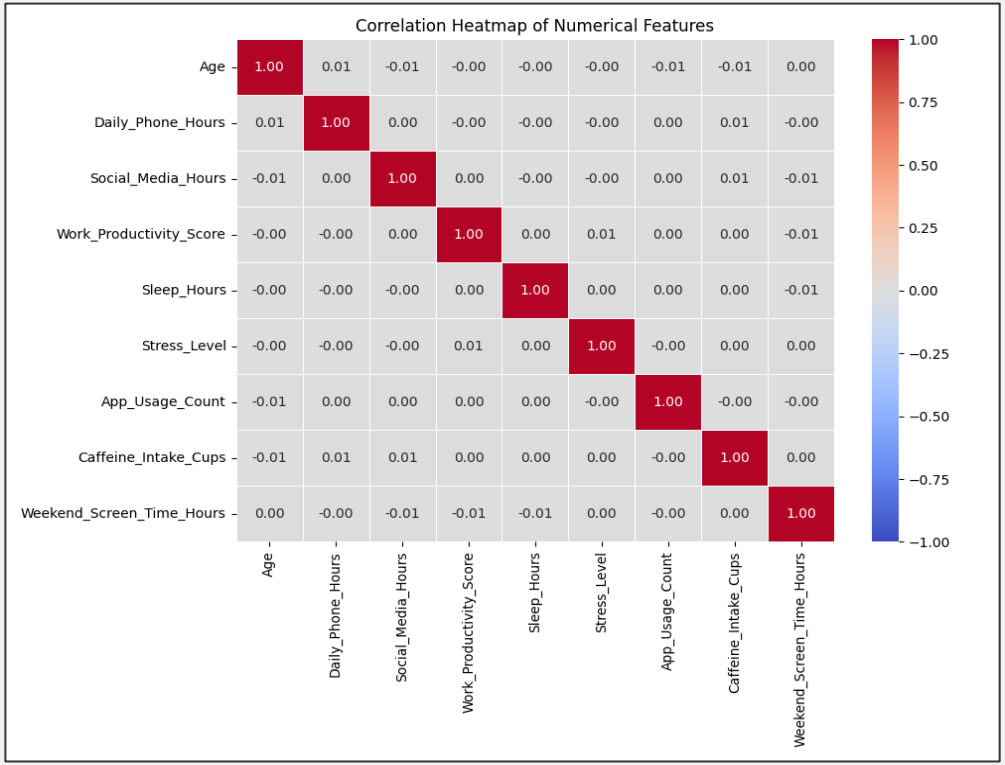


EDA 3 – Correlation Heatmap

- Shows relationship between variables.

Key Insights:

- More screen time ⟶ Less sleep.
- More social media ⟶ Lower productivity.
- Higher usage ⟶ Higher stress.



Machine Learning – Regression

Goal: Predict daily phone usage.

Model: Linear Regression.

Evaluation Metrics

Mean Squared Error (MSE).

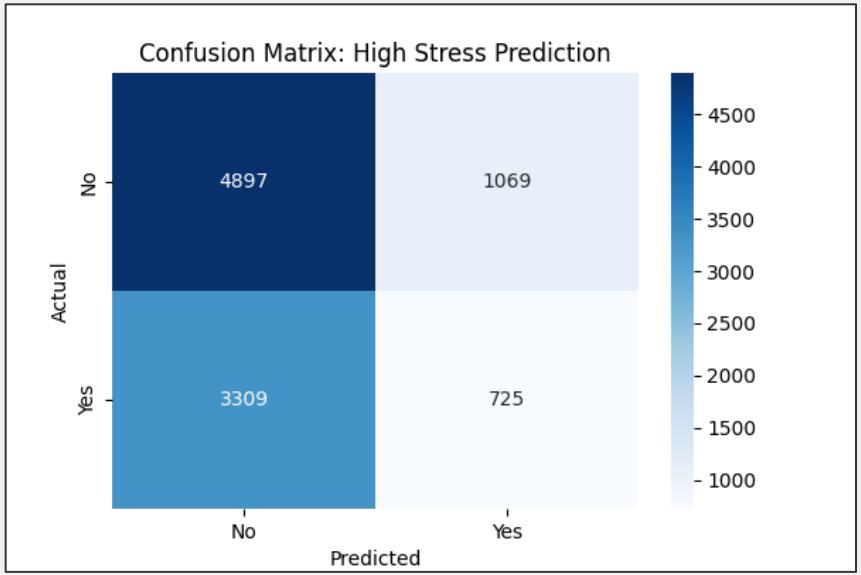
R^2 Score.

Machine Learning – Classification

Goal: Identify high stress users.

Model: Random Forest.

Output: Yes / No (High Stress)



Major Findings

High social media usage → Less sleep.

Weekend screen time is higher.

Heavy users show more stress.

Future Scope

Real-time screen tracking using apps.

Smart Alerts for:

- Reducing screen time.
- Improving mental health.

Integration with health monitoring systems.

Conclusion

- Screen time affects lifestyle and health.
- Data analysis helps understand user behavior.
- Can help people build healthier digital habits.

Thank You!